MGSC
MIDWEST GEOLOGICAL SEQUESTRATION CONSORTIUM

FAQs

Frequently Asked Questions

What is Geologic Carbon Dioxide Storage?

Geological storage of carbon dioxide (CO_2) is one method of isolating CO_2 from the earth's atmosphere. Reducing the amount of CO_2 released into the atmosphere may slow the global warming trends that have been observed in the last decade or more. Geological storage can play a significant role in preventing continued buildup of CO_2 in the atmosphere.

Carbon Capture, Utilization and Storage (CCUS) is the capture, transport, and storage of CO_2 from a point source, such as an electrical generation or biofuel production facility. The CO_2 is stored in the pore spaces of suitable geological formations in a porous rock, such as sandstone, and sealed in place with an impermeable caprock, such as shale. Utilization focuses on the use of CO_2 to produce additional fossil fuel resources, primarily through enhanced oil recovery.

Where is the Mount Simon Sandstone?

The Mt. Simon Sandstone lies more than one mile below Decatur, Illinois, and is 1,600 ft thick at this location. It is the deepest sandstone in the Illinois Basin, a bowl-shaped geological feature that holds thousands of feet of rock and underlies most of Illinois, southwestern Indiana, and western Kentucky. The Mt. Simon Sandstone is covered by the low-porosity Eau Claire Shale, which also has very low permeability (the ability of fluids to flow through a rock) and will therefore hold the CO_2 in the Mt. Simon. The Eau Claire Shale acts as a seal or cap, holding CO_2 deep in the Mt. Simon reservoir. The Mt. Simon has a large storage capacity because of its thickness, porosity, and areal extent, and it can hold a million tons of CO_2 within a radius of just about a quarter-mile from an injection well.







How is the CO_2 gas treated before injection at IBDP?

Carbon dioxide is being captured from the ADM ethanol production facility in Decatur, dehydrated, and transported via on-site pipeline to the storage site. The captured CO_2 is compressed to a liquid-like state so that it occupies less volume than CO_2 gas. The compressed CO_2 is then injected more than a mile below ground into porous sandstone called the Mt. Simon Sandstone for long-term geological storage.

What kinds of problems do you anticipate with having CO_2 in a liquid-like state?

We have not encountered any issues with the CO_2 in liquid state. We have drawn on the experience in the oil and gas industry with transporting and injecting supercritical CO_2. Using current technology, the pipelines and wells at IBDP have been engineered to handle the high pressures necessary to keep CO_2 in this form.